Q1 (1) - General Term and Middle term

Given, $(1+x)^m \cdot \left(1+\frac{1}{x}\right)^n$

To find Term independent of x

Coefficient of x° in $\left\{(1+x)^m \cdot \left(\frac{x+1}{x}\right)^n\right\}$

$\Rightarrow$ Coefficient of x° in $\left\{\frac{(1+x)^{m+n}}{x^n}\right\}$

$\Rightarrow$ Coefficient of x^n in $(1+x)^{m+n}$

$\Rightarrow$ Coefficient of x^n in $\{{}^{m+n}C_r x^r\}$

$\Rightarrow {}^{m+n}C_n$, put $r=n$

$\therefore$ Term independent of $x = {}^{m+n}C_n$ or ${}^{m+n}C_m$

Q2 (1) - General Term and Middle term

Given, $\left(x-\frac{1}{x}\right)^{2n+1}$

To find $(p+2)$ th term from the end of the expansion of $\left(x-\frac{1}{x}\right)^{2n+2}$

Now, $(p+2)$ th term from the end $=(2n+1-p-2+2)$ term from the beginning.

$(p+2)$ th term from the end $=(2n-p+1)$ th term from the beginning

[$\because$ In the binomial expansion of $(x+a)^n$, the r th term from the end is $(n-r+2)$ th term from the beginning]

$$\begin{aligned}
\therefore \quad T_{(2n-p+1)} &= {}^{2n+1}C_{(2n-p)}(x)^{(2n+1)-(2n-p)}\left(-\frac{1}{x}\right)^{2n-p} \\
&= (-1)^{2n-p} \cdot 2n+1C_{2n-p} \cdot x^{p+1} \cdot \frac{1}{x^{2n-p}} \\
&= (-1)^{2n-p} \cdot 2n+1 \\
&= (-1)^{2n-p} \cdot {}^{2n+1}C_{2n-p} \cdot x^{1+p-2n+p} \\
\text{or} \quad T_{(2n-p+1)} &= (-1)^p \cdot {}^{2n+1}C_{2n-p} \cdot x^{2p-2n+1} \\
&\left[\because (-1)^{2n-p} = (-1)^{2n} \cdot (-1)^{-p} = 1 \cdot \frac{1}{(-1)^p} = (-1)^p\right]
\end{aligned}$$

Q3 (2) - General Term and Middle term

Given, $\left(y^{1/5}+x^{1/10}\right)^{55}$

To find Number of terms free from radical sign.

Now, let $(r+1)$ th terms is independent of radical sign.

$\therefore \quad T_{r+1}={ }^{55} C_r\left\{(y)^{1/5}\right\}^{55-r}\left(x^{1/10}\right)^r$

$T_{r+1}={ }^{55} C_r y^{11-\frac{r}{5}} x^{\frac{r}{10}}$

Clearly, T_{r+1} will be independent of radical sign, if $r/5$ and $r/10$ are integers, where $0 \leq r \leq 55$.

$\therefore r=0,10,20,30,40,50$

Hence, there are 6 terms in the expansion of the given expression, which are independent of radical sign.

Q4 (3) - General Term and Middle term

Given, in the expansion of $(1+x)^m(1-x)^n$, coefficient of and x^2 are 3 and -6, respectively.

To find m in $(1+x)^m(1-x)^n$

We have, $(1+x)^m(1-x)^n=\left({ }^m C_0+{ }^m C_1 x+{ }^m C_2 x^2+\cdots+{ }^m C_m x^m\right)$
$\times\left({ }^n C_0-{ }^n C_1 x+{ }^n C_2 x^2+\cdots(-1)^{n n} C_n x^n\right)$

$={ }^m C_0{ }^n C_0-\left({ }^m C_0{ }^n C_1-{ }^n C_0{ }^m C_1\right) x$
$+\left({ }^m C_0{ }^n C_2+{ }^n C_0{ }^m C_2-{ }^m C_1{ }^n C_1\right) x^2+\ldots$

$\therefore$ Coefficient of $x=3$

$\Rightarrow \quad -\left({ }^m C_0{ }^n C_1-{ }^n C_0{ }^m C_1\right)=3$ ...(i)

Coefficient of $x^2=-6$

$\Rightarrow \quad \left({ }^m C_0{ }^n C_2+{ }^n C_0{ }^m C_2-{ }^m C_1{ }^n C_1\right)=-6$ ...(ii)

From Eq. (i), we have

$-(1 \cdot n-1 \cdot m)=3$

$\Rightarrow \quad m-n=3 \quad \ldots$ (iii)

From Eq. (ii), we have

$1 \cdot \frac{n(n-1)}{2!}+1 \cdot \frac{m(m-1)}{2!}-m n=-6$

$\Rightarrow \quad n(n-1)+m(m-1)-2 m n=-12$

$\Rightarrow \quad n^2-n+m^2-m-2 m n=-12$

$\Rightarrow \quad (m-n)^2-(m+n)=-12$

$\Rightarrow \quad (3)^2-(m+n)=-12 \quad$ [using Eq. (iii)]

$\Rightarrow \quad 9-(m+n)=-12 \quad \ldots$ (iv)

$\Rightarrow m+n=21$

On solving Eqs. (iii) and (iv), we get $m=12, n=9$

Q5 (2) - General Term and Middle term

Given, $\left(5^{1/2}+7^{1/8}\right)^{1024}$

To find Number of integral terms in the expansion of $\left(5^{1/2}+7^{1/8}\right)^{1024}$

The general term T_{r+1} in the expansion of $\left(5^{1/2}+7^{1/8}\right)^{1024}$ is given by

$T_{r+1}={}^{1024}\mathrm{C}_r\left(5^{1/2}\right)^{1024-r}\left(7^{1/8}\right)^r$

$T_{r+1}={}^{1024}C_r(5)^{512-r/2}\times 7^{r/8}$

$T_{r+1}={}^{1024}C_r(5)^{512-r+\frac{r}{2}}(7)^{r/8}$

$T_{r+1}={}^{1024}C_r(5)^{512-r}\left(5^{r/2}7^{r/8}\right)$

$T_{r+1}={}^{1024}C_r(5)^{512-r}\left(5^4\times 7\right)^{r/8}$

Clearly, T_{r+1} will be an integer, iff $r/8$ is an integer such that $0\leq r\leq 1024$

$\Rightarrow r$ is multiple of satisfying $0\leq r\leq 1024$.

$\Rightarrow \quad r=0,8,16,24,32,\ldots,1024$

Therefore, we have to find number of values of r.

$\therefore$ Let r have n values excluding '0'

i.e. $8,16,24,32,40,\ldots,1024$

This is an AP.

Here, $a=8, d=8$ and $I=1024$ $[\because I=a+(n-1)d]$

$\Rightarrow \quad 1024=8+(n-1)8$

$\Rightarrow \quad 1024=8+8n-8$

$\Rightarrow \quad 1024=8n$

$\Rightarrow \quad n=\frac{1024}{8}$

$n=128$

Including '0', r can assume 129 values.

Q6 (2) - Remainder problems

Let $E=2^{100}=2\cdot 2^{99}=2\cdot\left(2^3\right)^{33}=2\cdot(8)^{33}$
$=2\cdot(1+7)^{33}$

So, $E=2^{100}=2\cdot(1+7)^{33}=2\cdot\left(1+33\times 7+\frac{33\times 32}{2}(7)^2+\cdots\right)$

$$E=2+7\cdot 2\left(33+\frac{33\times 32}{2}\times 7+\cdots\right)$$

$$E=2+7k$$

Where, $k=2\cdot\left(33+\frac{33\times 32}{2}\times 7+\cdots\right)$

Now, $\frac{E}{7}=\frac{7k+2}{7}=k+\frac{2}{7}$

Hence, remainder is 2 .

Q7 (4) - Remainder problems

Here, $5^2=25$ which is close to $26=13\times 2$

Here, $E=5^{99}=5\times 5^{98}=5\times\left(5^2\right)^{49}=5(26-1)^{49}$

$\Rightarrow \quad E=5\left[{}^{49}C_0(26)^{49}-{}^{49}C_1(26)^{48}\right.$

$\left.+{}^{49}C_2(26)^{47}+\cdots{}^{49}C_{48}(26)-{}^{49}C_{49}\right]$

$\left[\because (x-a)^n={}^nC_0x^n-{}^nC_1x^{n-1}a+{}^nC_2x^{n-2}a^2+\cdots(-1)^n\tau_na^n\right]$

$\Rightarrow E=5\times 26\left[{}^{49}C_0(26)^{48}-{}^{49}C_1(26)^{47}+{}^{49}C_2(26)^{46}+\cdots{}^{49}C_{48}\right]-5$

$E=5\times 26k-5$

Where $k=\left[{}^{49}C_0(26)^{48}-{}^{49}C_1(26)^{47}+\cdots{}^{49}C_{48}\right]$

Now, $\frac{E}{13}=5\times\frac{26k}{13}-\frac{5}{13}=10k-\frac{5}{13}=10k-1+\frac{8}{13}$

Hence, the remainder is 8.

Q8 (1) - Remainder problems

We have, $(17)^{10}=\left(17^2\right)^5=(289)^5=(290-1)^5$

$(17)^{10}=(290-1)^5$

$={}^5C_0(290)^5-{}^5C_1(290)^4+{}^5C_2(290)^3-{}^5C_3(290)^2+{}^5C_4(290)-{}^5C_5$

$(17)^{10}=\left[(290)^5-5\times(290)^4\right.$

$\left.+10\times(290)^3-10\times(290)^2+5\times 290-1\right]$

$=\left[(290)^5-5\times(290)^4+10\times(290)^3-10\times(290)^2\right]+1450-1$

$=(290)^2\left[(290)^3-5\times(290)^2+10\times 290-10\right]+1449$

$=84100[24389000-420500+2900-10]+1449$

$=1000[841\times 2438900-841\times 42050+290-1]+1449$

$= 1000m + 1449$

Where , $m = [841 \times 2438900 - 841 \times 42050 + 290 - 1]$

Hence, last two digit in $(17)^{10}$ is 49.

Q9 (4) - Remainder problems

Given,

$N = 7^{100} - 3^{100}$

$N = (5+2)^{100} - (5-2)^{100}$

$\Rightarrow N = 2\left[{}^{100}C_1(5)^{99}(2) + {}^{100}C_3(5)^{97}(2)^3 + \cdots + {}^{100}C_{99}(5)(2)^{99}\right]$

$\left[\because (a+b)^n - (a-b)^n = 2\left({}^nC_1(a)^{n-1}b + {}^nC_3(a)^{n-3}(b)^3 + \cdots\right)\right]$

$\Rightarrow N = \left[{}^{100}C_1(5)^{99}(2)^2 + {}^{100}C_3(5)^{97}(2)^4 + \cdots\right] + 100 \times (5 \times 2) \times 2^{99}$

$N = k + 2^{99} \times 1000, \quad k \in I$

Hence, last three digit of the number is 000.

Q10 (1) - Properties of binomial coefficients based on sum

Sum of the given binomial coefficient is 256.

$\therefore 2^n = 256$

$\Rightarrow n = 8$

$\Rightarrow \left(2x + \frac{1}{x}\right)^8$

$$T_{r+1} = {}^8C_r(2x)^{8-r}\left(\frac{1}{x}\right)^r$$
$$= {}^8C_r 2^{8-r} x^{8-2r}$$

For constant term,

$8 - 2r = 0$

$\Rightarrow \quad r = 4$

Hence, constant term is ${}^8C_4 2^4 = \frac{8!}{4!4!} \times 16 = 1120$

Hence, option (1) is correct.

Q11 (2) - Properties of binomial coefficients based on sum

We have,

$\left(2x^2 - 3x + 1\right)^{11}$

Let $\left(2x^2-3x+1\right)^{11}=a_0+a_1x+a_2x^2+a_3x^3+\cdots$ ...(i)

Put $x=1$ in Eq. (i), we get

$0=a_0+a_1+a_2+a_3+\cdots$ ...(ii)

Put $x=-1$ in Eq. (i), we get

$6^{11}=a_0-a_1+a_2-a_3+\cdots$ ...(iii)

For the sum of even power of x, on adding Eqs. (ii) and (iii), we get

$$6^{11}=2\left(a_0+a_2+a_4+\cdots\right)$$

$$\Rightarrow a_0+a_2+a_4+\cdots=\frac{6\cdot 6^{10}}{2}=3\cdot 6^{10}$$

Hence, option (2) is correct.

Q12 (2) - Properties of binomial coefficients based on sum

We have,

$\left(1+x-3x^2\right)^{2145}=a_0+a_1x+a_2x^2+\cdots$

Put $x=-1$, we get

$a_0-a_1+a_2-a_3\cdots=(1-1-3)^{2145}=(-3)^{2145}$

$(-3)^{4\times 526}\times(-3)=(81)^{526}\times(-3)$

$=(1)^{526}(-3)$

$=-(\cdots 3)$

$\therefore$ End digit is 3.

Hence, option (2) is correct.

Q13 (3) - Properties of binomial coefficients based on sum

We have,

$(1+x)^n={}^nC_0+{}^nC_1x+{}^nC_2x^2+\cdots+{}^nC_nx^n$

$\Rightarrow \quad {}^nC_1{}^nC_0+2\frac{{}^nC_2}{{}^nC_1}+3\frac{{}^nC_3}{{}^nC_2}+\cdots+n\frac{{}^nC_n}{{}^nC_{n-1}}$

$\Rightarrow n+2\frac{(n-1)}{2}+3\frac{(n-2)}{3}-\frac{n(1)}{n}\left[\because \frac{{}^nC_r}{{}^nC_{r-1}}=\frac{n+1-r}{r}\right]$

$\Rightarrow n+(n-1)+(n-2)+(n-3)+\cdots+1$

$\Rightarrow \quad 1+2+3+4+\cdots+n$

$\Rightarrow \quad \frac{n(n+1)}{2}$

Hence, option (3) is correct.

Q14 (1) - Product of binomial coefficients and double summation

We have,

$$\left(x-\frac{{}^{50}C_1}{{}^{50}C_0}\right)\left(x-2^2\cdot\frac{{}^{50}C_2}{{}^{50}C_1}\right)\left(x-3^2\cdot\frac{{}^{50}C_3}{{}^{50}C_2}\right)\cdots\left(x-50^2\cdot\frac{{}^{50}C_{50}}{{}^{50}C_{49}}\right)$$

$$\Rightarrow x^{50}-x^{49}\left\{\frac{{}^{50}C_1}{{}^{50}C_0}+2^2\frac{{}^{50}C_2}{{}^{50}C_1}+3^2\frac{{}^{50}C_3}{{}^{50}C_2}+\cdots+50^2\frac{{}^{50}C_{50}}{{}^{50}C_{49}}\right\}+\ldots$$

$\therefore$ Coefficient of x^{49}

$$=-\left\{\frac{{}^{50}C_1}{{}^{50}C_0}+2^2\frac{{}^{50}C_2}{{}^{50}C_1}+3^2\frac{{}^{50}C_3}{{}^{50}C_2}+\cdots+50^2\frac{{}^{50}C_{50}}{{}^{50}C_{49}}\right\}$$

$$=-\left(\sum_{r=0}^{49}(r+1)^2\frac{{}^{50}C_{r+1}}{{}^{50}C_r}\right)=-\left(\sum_{r=0}^{49}(r+1)^2\frac{{}^{50}C_{r+1}}{{}^{50}C_r}\right)$$

$$=-\left[\sum_{r=0}^{49}(r+1)^2\frac{50!}{(r+1)!(50-r-1)!}\times\frac{r!(50-r)!}{50!}\right]$$

$$=-\left[\sum_{r=0}^{49}(r+1)^2\frac{(50-r)}{(r+1)}\right]=-\left[\sum_{r=0}^{49}(r+1)(50-r)\right]$$

$$=-\left[\sum_{r=0}^{49}\left(49r+50-r^2\right)\right]$$

$$=-\left[49\sum_{r=0}^{49}r+\sum_{r=0}^{49}50-\sum_{r=0}^{49}r^2\right]$$

$$=-\left[49\times\frac{49\times50}{2}+50\times50-\frac{49\times50\times99}{6}\right]$$

$$=-[49\times49\times25+2500-49\times25\times33]$$

$$=-[25\times49(49-33)+2500]$$

$$=-[25\times49\times16+2500]$$

$$=-22100$$

Hence, option (1) is correct

Q15 (3) - Product of binomial coefficients and double summation

Given sum $\sum_{i=0}^{m}\binom{10}{i}\binom{20}{m-i}$ where $\binom{p}{q}=0$, if $p<0$

To find Value of m, when the given sum is maximum

Now. $\sum_{i=0}^{m}{}^{10}C_i{}^{20}C_{m-i}$

$$={}^{10}C_0^{20}C_m+{}^{10}C_1^{20}C_{m-1}+{}^{10}C_2^{20}C_{m-2}+\cdots+{}^{10}C_m{}^{20}C_0$$

$=$ Coefficient of x^m in the expansion of product $(1+x)^{10}(1+x)^{20}$

$=$ Coefficient of x^m in the expansion of $(1+x)^{30}={}^{30}\mathrm{C}_m$

Now, ${}^{30}\mathrm{C}_m$ will be maximum, when $m=15$

Hence, option (3) is correct.

Q16 (3) - Product of binomial coefficients and double summation

Given, $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n \quad \ldots$ (i)

Put $x = 1$ in Eq. (i), we get

${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n$

$\Rightarrow \quad \sum_{r=0}^{n} C_r = 2^n$ ...(ii)

We have, $\sum_{r=0}^{n} \sum_{s=0}^{n} {}^nC_r \cdot {}^nC_s = \left(\sum_{r=0}^{n} {}^nC_r\right)\left(\sum_{s=0}^{n} C_s\right)$

$= \sum_{r=0}^{n} {}^nC_r \cdot 2^n$ [using Eq. (ii)]

$= 2^n \cdot 2^n$ [using Eq. (iii)]

$= (2)^{2n}$

Hence, option (3) is correct.

Q17 (1) - Product of binomial coefficients and double summation

Given,

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n \quad \ldots$ (i)

Put $x = 1$ in Eq. (i), we get

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n$

$$2^n = \sum_{r=0}^{n} C_r \quad \ldots \text{(ii)}$$

Now, $\sum_{r=0}^{n} \sum_{s=0}^{n} C_r C_s = \left(\sum_{r=0}^{n} C_r\right)\left(\sum_{s=0}^{n} C_s\right) = 2^n \cdot 2^n$

$\sum_{r=0}^{n} \sum_{s=0}^{n} C_r C_s = (2)^{2n}$ ...(iii)

We have, $\sum_{r=0}^{n} \sum_{s=0}^{n} C_r C_s = \left(\sum_{r=0}^{n} C_r^2\right) + 2\sum_{0 \le r < s \le n} C_r C_s$

$$\Rightarrow \quad (2)^{2n} = {}^{2n}C_n + +2\sum_{0 \le r < s \le n}\sum\sum C_r C_s$$

[using Eq. (iii) and $\sum_{r=0}^{n} C_r^2 = {}^{2n}C_n$]

$\Rightarrow \quad \sum\sum_{0 \le r < s \le n} C_r C_s = \frac{(2)^{2n} - {}^{2n}C_n}{2}$

Hence, option (1) is correct

Q18 (4) - Product of binomial coefficients and double summation

Given, $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$

We have,

$\sum_{r=0}^{n} \sum_{s=0}^{n} (C_r + C_s) = \sum_{r=0}^{n} (C_r + C_r) + 2\sum\sum\sum_{0 \le r < s \le n} (C_r + C_s)$

$(n+1)2^{n+1} = 2\sum_{i=0}^{n}(C_r) + 2\sum_{0\le r<s\le n}(C_r + C_s)$

$\Rightarrow \quad (n+1)2^{n+1} = 2\cdot 2^n + 2\sum\sum\sum_{0\le r<s\le n}(C_r + C_s)\ \left[\because \sum_{r=0}^{n} C_r = 2^n\right]$

$\Rightarrow (n+1)2^{n+1} - 2\cdot 2^n = 2\sum_{0\le 1<s\le n}\sum_{(C_r+C_s)}$

$\Rightarrow \quad n\cdot 2^{n+1} = 2\sum_{0\le r<s\le n}(C_r + C_s)$

$\Rightarrow \sum\sum_{0\le r<s\le n}(C_r + C_s) = n\cdot 2^n$

Hence, option (4) is correct.

Q19 (4) - Binomial theorem for any index

The expansion of $\frac{1}{(4-3x)^{1/2}}$ is valid iff $\left|\frac{3x}{4}\right| < 1$

$\Rightarrow |3x| < 4$

$|x| < \frac{4}{3}$

Hence, option (4) is correct.

Q20 (1) - Binomial theorem for any index

The general term of the expansion $(1-3x)^{\frac{-1}{3}}$ is

$\frac{\frac{-1}{3}\left(\frac{-1}{3}-1\right)\left(\frac{-1}{3}-2\right)\cdots\left(\frac{-1}{3}-(r-1)\right)}{r!}(-3x)^r$

$= \frac{(-1)^r\left[\frac{1}{3}\times\frac{4}{3}\times\frac{7}{3}\times\cdots\times\left(\frac{3r-2}{3}\right)\right]}{r!}(-1)^r 3^r x^r$

$\Rightarrow \quad \frac{1\cdot 4\cdot 7\cdots(3r-2)}{3^r\cdot r!}\cdot(3)^r x^r$

$\Rightarrow \quad \frac{1\cdot 4\cdot 7\cdots(3r-2)}{r!}x^r$

Hence, option (1) is correct.

Q21 (2) - Binomial theorem for any index

We have,

$\Rightarrow \left[1 + \frac{1}{2}\times 3x + \frac{(1+3x)^{1/2}(1-2x)^{-1/3}}{\frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\cdot(3x)^2}{2!}} + \cdots\right]$

$\left[1 + \frac{1}{3}(2x) + \frac{\frac{1}{3}\cdot\frac{4}{3}}{2!}(-2x)^2 - \cdots\right]$

$\Rightarrow$ Coefficient of x^2 is

$\frac{4\times4}{9\times2!}+\frac{1}{2}\times3\times\frac{1}{3}\times2+\frac{\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)}{2!}\times9=\frac{16}{18}+1-\frac{9}{8}$

$=\frac{64+72-81}{72}$

$=\frac{55}{72}$

Hence, option (2) is correct.

Q22 (1) - Binomial theorem for any index

The general term of the expansion $\left(\frac{1}{2}y^{1/3}+y^{-1/5}\right)^8$

$T_{r+1}={}^8C_r\left(\frac{1}{2}y^{1/3}\right)^{8-r}(y)^{-r/5}={}^8C_r\left(\frac{1}{2}\right)^{8-r}(y)^{\frac{8-r}{3}-\frac{r}{5}}$

The term is independent of y

if $\frac{8-r}{3}-\frac{r}{5}=0$

$\Rightarrow 40-5r-3r=0$

$\Rightarrow r=5$

$\therefore$ The term independent of y is 6th term

Hence, option (1) is correct.

Q23 (3) - Binomial theorem for any index

We have, $y=\frac{2}{5}+\frac{1\cdot3}{2!}\left(\frac{2}{5}\right)^2+\frac{1\cdot3\cdot5}{3!}\left(\frac{2}{5}\right)^3+\cdots$

$\Rightarrow \quad 1+y=1+\left(\frac{2}{5}\right)+\frac{1\cdot3}{2!}\left(\frac{2}{5}\right)^2+\frac{1\cdot3\cdot5}{3!}\left(\frac{2}{5}\right)^3+\cdots$

Let $\quad 1+y=(1+x)^n=1+nx+\frac{n(n-1)}{2!}x^2+\cdots$

On equating the given series, we get

$nx=\frac{2}{5}$ ...(i)

and $\frac{n(n-1)}{2}x^2=\frac{1\cdot3}{2!}\left(\frac{2}{5}\right)^2$ ...(ii)

On solving Eqs. (i) and (ii), we get

$$x=-\frac{4}{5}\text{ and }n=-\frac{1}{2}$$

$$\therefore 1+y=\left(1-\frac{4}{5}\right)^{-1/2}$$

$$\Rightarrow (y+1)=(5)^{1/2}$$

On squaring both sides, we get

$y^2 + 2y + 1 = 5$

$\Rightarrow y^2 + 2y - 4 = 0$

$\therefore$ The value of $y^2 + 2y - 4 = 0$

Hence, option (3) is correct.

Q24 (1) - Multinomial Theorem

We have, $(ab + bc + ca)^8$

The general term is $\frac{8!}{x!y!z!}(ab)^x(bc)^y(ca)^z$

$\Rightarrow \frac{8!}{x!y!z!}a^{x+z}b^{x+y}c^{y+z}$

Coefficient of $a^3b^4c^7$ in the above expansion is possible if

$x + z = 3, x + y = 4, y + z = 7$

$\therefore \quad x = 0, y = 4, z = 3$

$\therefore$ Coefficient of $(ab + bc + ca)^8$ is

$\frac{8!}{0!4!3!} = 280$

Hence, option (1) is correct.

Q25 (1) - Multinomial Theorem

We know that, the total number of terms in the expansion $(x_1 + x_2 + x_3 + \cdots + x_m)^n$ is ${}^{m+n-1}C_n$

$\therefore$ The number of terms in the expansion of $(a + b + c + d + e + f)^n$ is ${}^{6+n-1}C_n = {}^{n+5}C_n$

Hence, option (1) is correct.